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INTERNATIONAL ACADEMIC RESEARCH COMPETITION 2026
Research proposal on AquaGuard: A Federated Sense–Predict–Intervene Framework for Water Quality and Public Health Protection in Climate-Vulnerable Communities
Institution name: Sylhet Government Model School & College
Category: Junior
Country: Bangladesh
Submission date: 20 September, 2026
Researcher: Abdur Rahman Lahi.

Abstract

Climate-vulnerable communities face a persistent lag between the onset of water contamination and the recognition of resulting waterborne illness, because environmental monitoring and clinical case reporting operate as separate, disconnected systems. This proposal presents AquaGuard, an offline-first framework that fuses low-cost water-quality sensing with community-reported health symptoms collected through basic-phone USSD/SMS channels, processed through federated, on-device machine learning that shares only model updates rather than raw data. The framework is designed to flag statistically anomalous co-occurrences of water contamination and early symptom clusters before they are recognized by clinic-based surveillance, and to trigger both hyperlocal alerts and, where compatible hardware exists, automated treatment-dosing adjustments. This research proposes a pilot study to test whether such a sense-predict-intervene loop can shorten the time between contamination onset and community-level protective action, while preserving data ownership for participating communities.

1. Introduction / Research Problem

Unsafe water remains one of the most persistent global public-health burdens. As of 2022, 2.2 billion people still lacked safely managed drinking water, including 115 million people drinking untreated surface water [1], and this burden is concentrated in climate-vulnerable, low-resource regions where flooding, drought, and saline intrusion intensify contamination risk. Two response systems currently operate largely in isolation from one another. First, water-quality monitoring using Internet of Things (IoT) sensors and machine learning has advanced rapidly, but this body of work is almost entirely oriented toward predicting physicochemical parameters (turbidity, ppH, dissolved oxygen) rather than the human health outcomes that ultimately matter [2], [3]. Second, syndromic surveillance — the collection of self-reported symptoms to detect outbreaks earlier than laboratory-confirmed case reporting — has been piloted for waterborne disease, but a systematic review found no clear evidence that existing syndromic systems reliably improve early detection, in part because they are rarely combined with parallel environmental data streams [4]. Feasibility for low-connectivity health reporting has nonetheless been demonstrated: an SMS-based sentinel surveillance system deployed in Madagascar successfully used basic mobile technology for real-time outbreak reporting in a resource-constrained setting [5]. What has not yet been demonstrated is a system that fuses environmental contamination signals with community health-symptom signals in real time, on-device, in a way that is both privacy-preserving and capable of triggering an automatic protective response — closing the loop rather than only observing it.

2. Research Question(s)

Primary research question: Can a federated machine-learning model that jointly analyzes low-cost water-quality sensor data and basic-phone-reported community health symptoms detect waterborne contamination risk earlier than water-quality monitoring or clinical case reporting alone, while keeping raw data local to the community?

Sub-questions: (a) What lead time, in days, can be gained between an environmental contamination signal plus early symptom pattern and the point at which a waterborne outbreak

would otherwise be confirmed through clinic-based reporting? (b) How does a federated, differentially private model's detection accuracy compare to a centralized model trained on the same data? (c) Does an automatic alert-and-intervention trigger measurably shorten the time to a protective response (e.g., adjusted treatment dosing or a community advisory) compared to current practice?

3. Hypothesis

H1: A federated model that fuses time-aligned water-quality sensor readings with community-reported symptom counts will flag elevated waterborne-disease risk at least 3–5 days earlier, on average, than the date on which the corresponding cluster of cases would be confirmed through standard clinic-based reporting, without a statistically significant loss of detection accuracy relative to an equivalent centralized (non-federated) model.

4. Literature Review

Recent reviews confirm that machine learning and IoT-based water-quality monitoring is a mature and rapidly growing field, with random forest, support vector machine, and deep learning models (LSTM, CNN) achieving strong predictive accuracy for parameters such as turbidity, pH, and dissolved oxygen [2], [3]. These systems, however, are validated against water-quality benchmarks, not against human health outcomes, so a favorable sensor prediction does not by itself indicate whether anyone became ill. On the public-health side, syndromic surveillance has been explored specifically for waterborne outbreaks, but a systematic review covering studies from 1990–2018 found insufficient evidence that such systems reliably improve early detection when used on their own, and recommended future work explicitly investigate combining environmental and health data streams [4] — a gap this proposal targets directly. Separately, mobile-phone-based reporting has already proven operationally feasible in low-resource settings: Madagascar's SMS-based sentinel surveillance system demonstrated that encrypted, basic-phone data exchange can support real-time outbreak reporting without smartphone or broadband dependency [5]. Finally, federated learning has emerged as a privacy-preserving alternative to centralized health-data pooling, allowing multiple sites to train a shared model while keeping raw records local; a recent review summarizes the security and privacy benefits of this approach for healthcare applications and the threats it must still guard against, such as gradient-leakage attacks [6]. No study identified in this review, however, combines all four elements — low-cost environmental sensing, basic-phone syndromic reporting, federated on-device learning, and automated intervention — into a single closed loop for climate-vulnerable water systems. AquaGuard is positioned to fill that specific, previously unaddressed intersection.

5. Methodology

This is a design-and-pilot-validation study. (1) Site and partner selection: one climate-vulnerable community (coastal, flood-prone, or arid) will be selected in partnership with a local health clinic and water authority. (2) Sensing layer: solar-powered IoT pods measuring turbidity, conductivity, pH, and an E. coli proxy will be installed at shared water points, logging readings on a fixed interval and operating without continuous internet connectivity. (3) Health-reporting layer: a USSD/SMS short-code will allow residents and community health workers to report a short, structured list of symptoms (e.g., diarrhea, vomiting, fever) associated with waterborne illness,

timestamped and geotagged to the nearest water point. (4) Federated model: a lightweight anomaly-correlation model will run on a local edge device, learning the relationship between sensor-reading anomalies and lagged symptom-report spikes; only model weight updates — never raw sensor or symptom data — will be shared with a regional coordination hub, following the federated-averaging approach used in privacy-preserving healthcare ML [6]. (5) Intervention layer: when the correlated risk score crosses a pre-set threshold, the system will issue an SMS alert to households and clinic staff and, where a compatible automated chlorination/filtration unit is present, trigger a dosing adjustment. (6) Baseline comparison: the pilot will run in parallel with existing clinic-based case reporting to allow direct comparison of detection timing.

6. Data Analysis Plan

Sensor and symptom-report time series will be aligned to a common daily interval. Cross-correlation and Granger-causality analysis will be used to test whether contamination anomalies precede symptom-report spikes and by how many days, directly informing the lead-time component of H1. Model performance will be reported using precision, recall, and F1-score for correctly flagged risk periods, benchmarked against clinic-confirmed case clusters as ground truth. The federated model's detection accuracy will be compared against a centralized model trained on pooled data from the same pilot to quantify any privacy-utility trade-off, and the differential-privacy noise parameter will be varied to characterize this trade-off explicitly. All comparisons will use a held-out validation window not seen during model training.

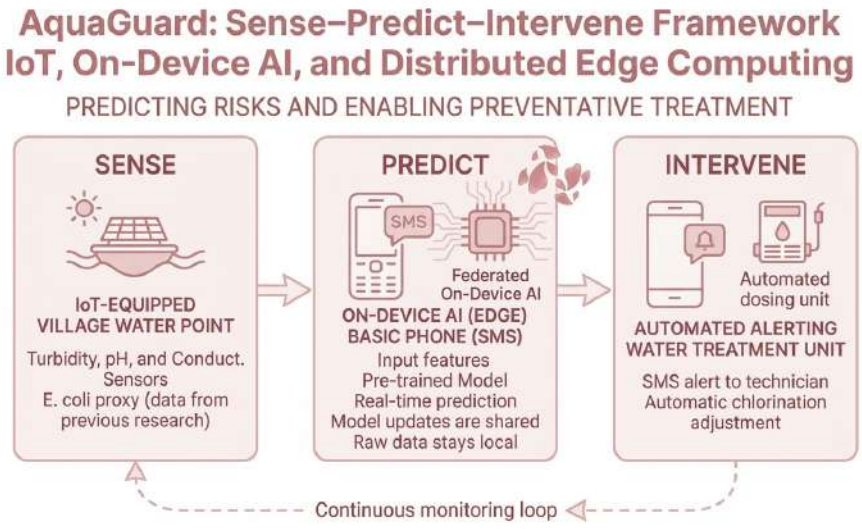


Figure 1. AquaGuard: Sense–Predict–Intervene Framework

7. Timeline

Phase	Activity	Duration
1	Finalize literature synthesis, identify pilot community and clinic/water-	Weeks 1–3
2	Community consent, sensor procurement, USSD short-code setup	Weeks 4–6
3	Sensor and reporting-channel deployment at pilot site	Weeks 7–8
4	Data collection (sensor + symptom reports) running in parallel with existing clinic reporting	Weeks 9–16
5	Federated model training, validation, and privacy-utility analysis	Weeks 17–20
6	Results synthesis, write-up, and preparation of finalist materials	Weeks 21–22

8. Conclusion

Existing water-quality AI systems predict contamination; existing syndromic surveillance systems observe symptoms; neither closes the loop by connecting the two in real time or by acting on the correlation automatically. AquaGuard proposes to test whether a federated, privacy-preserving model that fuses low-cost environmental sensing with basic-phone community health reporting can detect waterborne-disease risk earlier than either data stream alone, and whether an automatic alert-and-intervention step can shorten the time to protective action for climate-vulnerable communities. If validated, this pilot would offer a low-cost, replicable, and data-sovereignty-preserving model for water-related public-health protection in regions currently underserved by both water-quality monitoring and health surveillance infrastructure.

References

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